

Laplace transforms of standard functions

1. Prove that $L\{t^n\} = \frac{n!}{s^{n+1}}$ where $n = 0, 1, 2, 3, \dots$

2. Find Laplace transform of

(a) $\sin^3 t$

(b) $\cos(3t)\sin^2 t$

3. Find Laplace transform of $\cos^3 2t$

(a) $\frac{6}{(s^2 + 1)(s^2 + 9)}$

ans:

(b) $\frac{2s(s^2 - 23)}{(s^2 + 1)(s^2 + 9)(s^2 + 25)}$

ans: $\frac{s(s^2 + 28)}{(s^2 + 36)(s^2 + 4)}$